

Use-Case Flow: Compute Debt Settlement

Exception Flow

Trip Expense Simplification Engine — Problem Statement #37

Purpose

Exception flows describe error conditions that prevent the “Compute Debt Settlement” use case from completing normally, as distinct from alternate flows, which describe valid variations still leading to a successful outcome. Each exception below ties back to a functional or non-functional requirement in the Requirements Table and specifies the system's recovery behavior.

Common Postcondition

For every exception listed, the failure postcondition already defined for the use case applies: no settlement graph is generated (or a partially-applied one is rolled back), and previously recorded balances remain unchanged unless explicitly noted otherwise.

Exception Catalog

ID	Exception	Triggering Condition	System Response
EX-1	System Failure During Min-Flow Computation	The settlement algorithm (FR-001) crashes, times out, or exceeds the 100 ms / 50-traveler / 500-expense budget defined in NFR-001 while computing the settlement graph.	System aborts the computation, leaves all existing balances unchanged, logs the failure, and displays an error message asking the Group Traveler to retry. No settlement edges are created.
EX-2	Currency Service Unreachable With No Cached Rate	One or more expenses are in a foreign currency and no cached exchange rate exists at all (first-time currency, cache expired), so FR-003 cannot normalize the expense to the base currency.	System halts settlement computation, flags the specific expenses that could not be normalized, and displays an error explaining that settlement cannot proceed until those expenses are converted. Existing balances remain unchanged.
EX-3	Concurrent Balance Modification	While the Group Traveler is reviewing the settlement plan, another traveler or the Trip Admin adds/edits an expense or removes a traveler (FR-002 / FR-004),	System detects the version conflict when the Group Traveler attempts to confirm a payment, discards the stale settlement plan, notifies the Group Traveler that balances

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		invalidating the balances the plan was computed from.	changed, and requires the plan to be recomputed (returns to step 2 of the Main Success Scenario).
EX-4	Confirmed Payment Cannot Be Persisted	The system accepts a Group Traveler's confirmation of a settlement edge (step 6) but fails to durably record it, e.g. due to a database or network error.	System rolls back the edge to "unpaid," displays an error indicating the confirmation was not saved, and instructs the Group Traveler to retry. The traveler's net balance is not altered.
EX-5	Encryption / Security Control Failure	The system cannot satisfy NFR-002 (encryption of stored/transmitted financial data) for the balances or payment details involved in the settlement, e.g. a TLS handshake failure or key-management error.	System refuses to display or transmit the affected balances or settlement plan, logs a security event, and shows a generic error to the Group Traveler rather than exposing partial financial data.